



How warm-versus competent-toned AI apologies affect trust and forgiveness through emotions and perceived sincerity

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ABSTRACT

As generative artificial intelligence (GenAI) becomes more integrated into corporate communication, its role in crisis messaging raises critical questions about audience perception and trust. Drawing on theories of machine heuristics, this study explores how relational cues in AI-authored crisis apologies shape emotional and cognitive responses that ultimately influence trust and forgiveness. A 3 (authorship attribution: AI vs. human vs. control) x 2 (relational tone: warmth vs. competence) between-subjects factorial design with 464 participants was conducted to assess if and how incorporating a warm tone into AI-generated apologies can help overcome AI's inherent limitations associated with machine heuristics. Results show that human-authored apologies are perceived as more sincere, with warmth enhancing their positive impact. AI authorship elicited more negative emotions and reduced perceived sincerity compared to human authorship; however, relational tone was found to moderate the indirect effects of authorship on trust and forgiveness through negative emotions and perceived sincerity. These findings highlight the importance of both emotional and cognitive mechanisms in AI-mediated communication. This research advances an understanding of AI-mediated communication, identifying relational tone as a critical moderator of machine heuristic effects in crisis communication contexts. By integrating both emotional (negative affect) and cognitive (perceived sincerity) mediators into the model, this research provides a deeper understanding of how audiences evaluate and respond to AI-generated apologies in crisis contexts. Additionally, it offers a novel application of machine heuristic theory, extending its relevance to reputational management and organizational transparency.

1. Introduction

The rise of generative artificial intelligence (GenAI) is rapidly transforming how organizations manage crisis communication. GenAI supports all stages of crisis response, from early planning to post-crisis communication, by enabling rapid content creation and low-risk message testing (Karinshak, 2024). During an actual crisis, GenAI assists practitioners operating under time constraints by facilitating quick development and adaptation of messaging addressing stakeholders' concerns (Karinshak, 2024). These capabilities offer strategic advantages, particularly when organizations must respond swiftly and credibly across digital platforms. Despite these advantages, the use of GenAI in crafting crisis apologies remains contentious. Prior research indicates that AI-attributed apologies are perceived less effective than human-authored ones in restoring trust and eliciting forgiveness, largely due to diminished perceptions of sincerity, authenticity, and empathy (Glikson & Asscher, 2023; Lim et al., 2025). However, as Large

Language Models (LLMs) continue to advance, the quality and emotional resonance of GenAI-generated responses have also improved. Reflecting on this progress, Ovsyannikova et al. (2025) demonstrate that AI-generated responses to hypothetical, emotionally charged scenarios (e.g., personal loss or distress) were rated as more compassionate and responsive than human-written messages across four empathy-focused studies.

Given such rapid progress, it is increasingly impractical to assume that organizations can avoid using AI in high-pressure crisis situations that require prompt and tailored responses. This growing reliance on GenAI for crisis communication raises a critical question: Can AI-generated apologies—whether explicitly disclosed or inferred through detection tools—achieve essential crisis communication outcomes such as regaining trust and eliciting forgiveness? Considering that GenAI systems now produce responses that closely emulate human tone, rhetorical style, and contextual sensitivity with remarkable accuracy (Baron, 2023; Harari, 2024), this question has become increasingly

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warranted.

In addressing the research inquiry, we identified two relevant streams of research. The first focuses on how individuals evaluate an organization's crisis communication when they learn that the message was generated or assisted by generative AI. This includes both case studies (Piller, 2025) and experimental studies (Glikson & Asscher, 2023; Lim et al., 2025). Findings from this line of research show that apologies attributed to AI are often perceived as less effective in fostering trust and forgiveness than identical messages attributed to human sources. The second stream examines whether the relatively weaker impact of AI-authored crisis communication can be moderated by the tone or style of the message (Huang & Ki, 2024; Oh & Ki, 2024; Xiao & Yu, 2025).

However, neither stream explains the psychological mechanisms through which an AI-generated apology might lead to positive outcomes based on its tone, particularly when it conveys competence or warmth. Building on the Stereotype Content Model (Fiske et al., 2007), which posits that perceptions of others are primarily based on warmth and competence, we propose that messages emphasizing these traits may lead to differing evaluations of AI-authored crisis responses. Warmth cues may help offset AI's perceived deficits in empathy and sincerity. In this context, we identify empathy and sincerity as key mediators that could reduce negative reactions and enhance perceived credibility of AI-generated apologies. Despite their importance, no known research has examined the interaction effects of message authorship and relational tone on trust and forgiveness, nor explored the mediating roles of negative emotion and perceived sincerity.

To fill this gap, this study examines two distinct, theory-driven pathways through which an identical apology statement, attributed to either GenAI or a human source, may lead to trust and forgiveness intentions by reducing negative emotions and increasing perceived sincerity. It also examines how these indirect effects are moderated by relational tone. By examining these mediating and moderating processes, this study advances the theoretical understanding of how AI-generated apologies can be strategically crafted to minimize negative perceptions and foster trust and forgiveness.

2. Literature review

2.1. From algorithmic to intelligent content generation

AI-driven content generation has become a major technological innovation across communication sectors, including journalism (Moravec et al., 2024), sports broadcasting (Geissler et al., 2024; Jung et al., 2025), over-the-top media services (Shin et al., 2024), and social media (Brüns & Meißner, 2024). Prior to the launch of ChatGPT by OpenAI in November 2022, algorithmic content generation, particularly in the form of automated journalism or "robot reporters," represented a major innovation in digital newsrooms. Prominent news organizations, including The Associated Press, The Wall Street Journal, and The New York Times, have employed AI technologies to generate structured news reports in areas, such as finance, weather, and sports using rule-based, data-driven algorithms (Marconi, 2020). However, these systems relied heavily on template-based reporting and rule-based algorithms, which resulted in outputs often described as "cookie-cutter-like" and lacking "personal writing voice" rich in human-authored writing (Baron, 2023, p. 223).

Recent developments have significantly expanded the role of AI in journalism. Simon (2025), based on interviews with 34 journalists and editors across major newsrooms, found that AI now contributes to multiple aspects of editorial workflows. These include content generation, trend analysis, audience segmentation, source verification, and headline and visual optimization. This reflects a broader integration of AI into the core stages of news production. Pavlik (2023) described a similar evolution toward hybrid collaborative models in which AI tools such as expert systems and natural language processing support human

journalists. These tools have enhanced reporting accuracy and speed, especially in data-intensive or multilingual environments. These technological advancements reflect a transition from basic automation to more context-aware, adaptive AI systems that contribute to content creation with increasing editorial sophistication. Similarly, the emergence of GenAI has transformed the business of AI-driven content. GenAI systems, capable of generating not only functional prose but also poems, stories, and long-form narratives that are difficult to distinguish from human writing (Baron, 2023), now produce content that mimics human language with striking fluency. As AI-generated content becomes more integrated into various tasks and professional settings, scholars have increasingly turned their attention to the use of AI in sensitive contexts, such as crisis communication (Xiao & Yu, 2025).

As it becomes both technically feasible and practically achievable to produce AI-generated content that matches the quality of human writing, a growing area of interest centers on the effectiveness of AI-authored apology statements in managing public perception during corporate or personal crises. Of particular interest is how the public responds when the use of AI in crafting an apology is either voluntarily disclosed or later revealed by others. The key question is whether an AI-generated or AI-assisted apology, when equal in quality to a human-written one, can produce comparable communication outcomes, such as restoring trust and credibility.

This question hinges on evolving social norms that guide perceptions of task appropriateness when performed by humans versus machines. These norms are not static but shift over time, often in response to technological advancement. For example, autonomous driving has become more widely accepted today than it was a decade ago. In emotionally sensitive or morally charged domains, such as delivering apologies, perceptions of appropriateness often depend on whether the public views AI as not only technically capable but also sincere and empathetic in expressing remorse. Many individuals may question AI's ability to convey emotional depth, relying on what is known as the machine heuristic—a mental shortcut through which people form judgments based on their understanding of machines as social actors (Yang & Sundar, 2024). Importantly, this heuristic is not inherently negative. While some tasks, particularly those involving logic or operational efficiency (Lazaroiu & Rogalska, 2023), may be deemed well-suited to AI, others that require human accountability, emotion, or ethical judgment may be seen as less appropriate when associated with machine-generated content (Pelau et al., 2021). Thus, AI systems are generally accepted in roles requiring analytical or technical capabilities, but they face resistance in contexts demanding high levels of empathy or ethical discernment, such as organizational apologies. For instance, when Vanderbilt University disclosed the use of ChatGPT to draft a condolence message following a campus tragedy, the reaction was overwhelmingly negative, with many interpreting the AI involvement as a lack of sincerity and rhetorical humanity (Piller, 2025). Similar concerns are echoed in experimental work showing that AI-mediated communication, especially in interpersonal contexts like apologies, is often perceived as less authentic and elicits lower willingness to forgive unless efforts are made to limit AI involvement or convey genuine human intent (Glikson & Asscher, 2023).

These perceptions of appropriateness and trust often depend on how individuals evaluate an AI agent's competence and its ability to avoid errors. Xiao and Yu (2025) found that stakeholders' trust in AI agents hinges largely on perceived competence, which in turn affects satisfaction and attributions of responsibility. Their study supports the view that machine heuristics are not applied uniformly but vary depending on the context. Further extending research on judgment heuristics, Dietvorst et al. (2015) identified a phenomenon known as algorithm aversion, in which individuals tend to reject algorithmic or data-driven decision-making even when such systems consistently outperform human judgment. This reluctance becomes more evident after the algorithm makes an error; individuals are more likely to lose trust in a machine and less willing to rely on it again compared to a human who makes a similar

mistake. An earlier study also found that individuals tended to be more sensitive to errors made by automated systems than those made by humans (Madhavan & Wiegmann, 2007). This asymmetry reflects a deep cognitive bias in how people evaluate competence and reliability across human versus machine agents.

2.2. Authorship attribution in crisis communication

In the AI era, a central issue in the communication industry is whether the work of human practitioners can or will be replaced by GenAI systems (Xiao & Yu, 2025). Many in the field view this development as inevitable, particularly once the use of LLMs becomes normalized and embedded into everyday workflow (Liang et al., 2025; Przegalinska et al., 2025). Liang et al. (2025) found that, following the release of ChatGPT, as many as 24 % of corporate press releases published by late 2024 were generated or assisted by LLMs. This trend extended beyond the private sector. Nearly 14 % of United Nations press releases were also found to have been generated or modified using LLMs. Once such a shift occurs, it may be difficult to reverse. This brings us to two key questions: First, can GenAI produce crisis communication that rivals the quality of those crafted by experienced communication professionals? Second, will the public perceive a GenAI-generated crisis message, particularly an apology, as a legitimate and equally acceptable form of organizational response?

Answering these questions requires empirical research that spans diverse crisis contexts and examines how authorship and tone influence emotional and perceptual outcomes in AI-mediated communication. A few recent studies and cases offer early insights into how these dynamics play out in practice. A notable example is Vanderbilt University's Peabody Office of Equity, Diversity, and Inclusion (EDI), which issued a message in response to a mass shooting at Michigan State University. The message conveyed a sense of community, empathy, and collective responsibility to create a supportive environment; however, at the end of the message, it was disclosed in small text that it was paraphrased using ChatGPT (Piller, 2025). The backlash was swift, with stakeholders, especially students, expressing moral outrage that underscored an assumption for authenticity during emotionally charged, high-severity crises. Critics argued that the use of AI in such a sensitive context lacked empathy and violated expectations of human connection, particularly by outsourcing the emotional labor of crisis messaging to AI (Piller, 2025). Shortly after the incident, the Dean of Peabody College issued a formal apology acknowledging that the message had "missed the crucial need for personal connection and empathy during a time of tragedy" (Wu, 2023). Reflecting on this case, Piller (2025) argues that in crisis communication, how a message is composed can overshadow what it says. He emphasized the importance of "rhetorical humanity," which refers to the perceived effort, care, and ethical responsibility that audiences expect from communicators during emotionally charged crises.

Recent studies in crisis communication have examined the role of authorship attribution in crisis communication outcomes. Kim and Song (2021) conducted research on AI-mediated crisis communication by examining how the presentation of AI agents, as either human-like or machine-like, moderates the effectiveness of apology attribution strategies. They found that machine-like agents were more effective when they issued external apologies that deflected blame, while human-like agents were more effective when they offered internal apologies accepting responsibility. Lim et al. (2025) investigated how identical corporate apologies attributed to either a human or an AI agent influenced public trust and forgiveness in the context of a data breach. They found that statements attributed to a human source generated significantly higher levels of trust and forgiveness intentions than those attributed to AI. Their mediation analysis further revealed distinct underlying mechanisms: the positive effect of human-attributed statements was mediated by perceived empathy, whereas the negative effect of AI-attributed statements was mediated by reduced perceptions of sincerity. Based on previous research, we assumed that authorship

attribution will significantly influence perceived sincerity, negative emotions, trust in the organization, and forgiveness intentions. Thus, we posit the following hypotheses.

H1. Apologies attributed to a human author will generate significantly (H1a) higher perceived sincerity, (H1b) lower negative emotional responses, (H1c) greater trust in the company, and (H1d) greater forgiveness intentions than those attributed to an AI author.

2.3. Relations tone in crisis communication

Salzano and Ashby-King (2025) found current discussions around GenAI in public relations often emphasize efficiency and productivity, while downplaying the field's emphasis on relational and dialogic communication. Crisis communication scholars have emphasized that effective responses must go beyond being prompt and transparent and that they must also show empathy (Schoofs et al., 2022) and sincerity (Stephens et al., 2019) to engage the public meaningfully. Through interviews with communication professionals across Europe, Schoofs and Frances (2022) reveal that empathy is consistently regarded as a non-negotiable element of effective crisis messaging. Practitioners emphasized that an empathetic tone is critical for restoring trust, calming public anger and outrage, and humanizing the organization's crisis response during moments of reputational threat. Given the key role of empathy in crisis communication, a critical question emerges in the context of generative AI: Even if AI can mimic a human-like tone, can an apology attributed to AI convey the same level of empathy and warmth typically expected from human communicators? The severe backlash following Vanderbilt University's Peabody Office of EDI's disclosure of using ChatGPT highlights concerns about whether AI-generated apologies, whether disclosed or detected, can effectively achieve the goals of crisis communication, such as restoring trust and conveying genuine concern.

The potential negative association between AI and content quality can be explained using the concept of machine heuristics, which refers to cognitive shortcuts triggered by cues such as authorship labels or perceived agency (Yang & Sundar, 2024). While machine heuristics can lead to both positive and negative evaluations or judgments, people tend to associate AI with stereotypical traits such as expert, efficient, and rule-based (Yang & Sundar, 2024). These traits align more closely with the competence dimension as defined in the Stereotype Content Model (Fiske et al., 2007). Competence includes perceptions of capability, skill, confidence, and intelligence. In contrast, warmth encompasses traits such as sincerity, friendliness, helpfulness, and moral intention. Warmth, which has a broader definition than empathy, focuses on the ability to foster positive relationships through genuine and supportive interactions (Cuddy et al., 2011; Wang et al., 2025). These warmth-related qualities may be expected in human-to-human crisis communication, particularly in apology contexts where empathy and trustworthiness are essential to rebuild damaged relationships. Oh and Ki (2024) also found that conversational human voice, which reflects a warm relational tone, increased organization-public relationship (OPR) outcomes regardless of whether the communicator was human or AI. Wang et al. (2025) examined how warmth in AI-generated responses influences customer intentions to reuse a service following service recovery. They found that warm-toned AI responses significantly increased reuse intentions by enhancing perceived social presence, even when the responses were perceived as emotionally simulated rather than genuinely human. They discussed that warmth could act as a social cue that compensates for AI's lack of true emotional capacity, such that warm-toned AI messaging may help alleviate negative emotions and improve relational outcomes, even in the absence of authentic emotional expression. These findings support the growing consensus that tone matters as much as authorship (Oh & Ki, 2024; Xiao & Yu, 2025), especially when emotional reassurance and the perception of sincerity are communication goals. Based on these insights, we assume that

relational tone will significantly influence perceived sincerity and negative emotions. Thus, we posit the following hypothesis.

H2. Warm-toned apologies will elicit (H2a) higher sincerity perceptions and (H2b) lower negative emotions than competence-toned apologies.

2.4. Relational Tone's moderating role

Finding the right tone for apologies during organizational scandals or crises has long been a primary concern for both practitioners and scholars, as the tone of the message can influence stakeholder perceptions by affecting emotional responses (ten Brinke & Adams, 2015).

Perceptions of warmth and competence are central to how audiences form social judgments about communicators, whether they are human or AI. These judgments, in turn, influence not only emotional responses such as trust or skepticism but also behavioral outcomes, including engagement, forgiveness, and support (Cuddy et al., 2011). Warmth and competence are important in influencing stakeholder responses during crises, particularly when assessing corporate culpability or product-related failures. Research in crisis communication suggests that effective messaging often involves balancing both competence and warmth, depending on the type of crisis and the audience's expectations of the communicator (Pei, 2025). For example, warmth may be especially important in crises involving ethical or moral failures, while competence may be more critical in situations related to technical breakdowns or product safety.

Research has shown that warmth-focused crisis responses can foster forgiveness and rebuild trust by mitigating negative attributions (Barbarossa et al., 2018; Xie & Peng, 2009). In contrast, competence-focused messaging highlights the organization's ability to manage the crisis effectively and prevent recurrence, which becomes especially effective in product-harm or operational crises (Jeong, 2024). Applying these perceptions to influencer marketing, Roccapiore and Pollock (2023) found that warmth cues were more effective in fostering deeper, cognitively effortful engagement, such as ongoing interaction or support, whereas competence cues were more effective in generating quicker, low-effort behaviors, such as following or superficial engagement.

Recent studies have begun to examine how the relational tone of warmth versus competence interacts with the perceived identity of the communicator, attributed to a human or an AI (Oh & Ki, 2024) or to humanoid versus non-humanoid agents (Choi et al., 2021). In related research, Kim and Song (2021) investigated the interaction between anthropomorphism (human-like vs. machine-like AI agents) and responsibility attribution (internal vs. external) in trust repair. A central question in these studies is whether non-human agents, such as generative AI or non-humanoid robots, can compensate for their perceived lack of emotional capacity in terms of warmth and empathy, when delivering an apology. We assume that in such contexts, relational tone or additional human engagement may help AI agent make up for their inherent limitations stemming from negative machine heuristics. Choi et al. (2021) noted that non-humanoid agents are perceived as less able to "convey empathy and sincerity" (p. 358), and therefore, their apologies may require additional human-like cues or engagement to be effective.

In evolving research applications, some researchers extended the relative efficacy of warmth and competence in investigating AI-mediated crisis communication (e.g., Hu & Pan, 2024; Xiao & Yu, 2025). Xiao and Yu (2025) examined how perceptions of competence and emotional support (as a proxy for warmth) influence stakeholder responses in AI-mediated crisis communication. They found that chatbots delivering competence-focused, instructional messages led to higher satisfaction and lower responsibility attribution than those offering empathy-based, adjusting messages. In contrast, warmth-oriented responses were no more effective than instructing messages and did not

outperform human agents. Their findings suggest that chatbots' limited emotional capacity weakens the impact of warmth-based communication. Hu and Pan (2024) investigated how informal language styles, particularly those conveying cuteness and social warmth, influence customer forgiveness following service failures by AI-powered conversational agents. Their study demonstrated that using cute or informal language significantly increased customers' willingness to forgive AI agents, with this effect mediated by enhanced perceptions of warmth. Furthermore, the indirect effect of these language cues was moderated by relationship norms. Their findings suggest that although AI agents typically lack genuine emotional capacity, the use of warmth-evoking tones and styles can help offset this relational deficit, especially when the tone aligns with the audience's relationship expectations. Together, these studies suggest that while simulated warmth cues can help humanize AI, their impact is limited, context-dependent, and rarely matches the effectiveness of human-delivered warmth.

Drawing from aforementioned previous research, we expected that higher perceived sincerity would primarily ensue in the human-attributed apology condition. However, we also predicted that, within the AI-attributed condition, a warm tone would be more effective than a competent tone in reducing negative emotional responses. Thus, we posit the following hypotheses.

H3. Warm-toned apologies will reduce negative emotions more effectively in AI-attributed apologies than in human-attributed apologies.

H4. Warm-toned apologies will enhance perceived sincerity more strongly in human-attributed apologies than in AI-attributed apologies.

2.5. Relational tone as a moderator of indirect effects through emotion and sincerity

While prior studies have examined the individual effects of authorship attribution, machine heuristics, and relational tone in crisis communication, recent research suggests a theoretical mechanism by which machine heuristics triggered by authorship interact with relational tone to influence crisis response outcomes through their impact on emotions and perceived sincerity. Cao et al. (2024) found that anthropomorphic cues in chatbots, such as warm language or human-like design, reduced users' perceptions of error severity and emotional resistance following a service failure, which in turn led to greater forgiveness. Pelau et al. (2021) found that empathetic interaction quality and perceived psychological anthropomorphism (e.g., having a friendly voice) increased user acceptance of AI in service contexts. In crisis communication, relational tones that convey warmth (e.g., empathy) or competence (e.g., problem-solving ability) can reduce negative emotions and improve the perceived sincerity of an apology. Applying this to AI-authored messages, warmth cues that suggest human-like qualities may ease emotional resistance and support greater forgiveness and trust restoration. In a study on linguistic style in AI service failure, Hu and Pan (2024) found that informal, warmth-oriented language increased forgiveness by increasing mind perception, or the belief that the AI agent had emotional and intentional capacities. This belief led to greater perceptions of warmth and sincerity, which reduced negative emotions and increased the willingness to forgive. Although their study focused on direct effects, the results support the view that relational tone influences forgiveness through both emotional and cognitive responses. In the context of AI-authored crisis messages, language that conveys warmth or competence can affect how recipients respond emotionally and how sincere they judge the message to be. Taken together, these studies suggest that relational tone may influence how machine heuristics and anthropomorphic perceptions affect users' forgiveness intentions. Specifically, the reviewed findings point to both emotional (e.g., reduced negative feelings) and cognitive (e.g., greater perceived sincerity) mechanisms as central to building trust and forgiveness intentions in AI-driven crisis communication.

Jeong (2024) further emphasized the importance of aligning the message tone with the crisis type. She showed that warmth is more effective in moral-harm crises, while competence is more effective in product-harm scenarios. These findings suggest that relational tone can amplify or buffer perceptions such as sincerity or artificiality, depending on how the communicator is socially perceived. Yu and Zhao (2024) also provided direct evidence of moderated mediation. They found that the effect of warmth cues (e.g., emojis) on satisfaction through perceived warmth was stronger when the chatbot was less autonomous. Their results further support the idea that tone can buffer the relational limitations of AI. Wang et al. (2025) also found that the effectiveness of these cues, such as emojis, depends on the communication style used, such as that informal, warm-toned messages can amplify the positive impact of emojis on perceived sincerity and trust. Xiao and Yu (2025) identified competence as AI's comparative strength yet found that chatbots lacking warmth performed worse than humans in apology contexts. Their findings suggest that relational tone may help offset AI's limitations in conveying emotional depth.

Taken together, these studies suggest that relational tone (warmth vs. competence), when interacting with authorship attribution (AI vs. human), can produce distinct emotional and perceptual responses, which in turn influence crisis communication outcomes. Specifically, we propose that relational tone moderates the indirect effects of authorship on trust and forgiveness through mediators of negative emotions and perceived sincerity.

H5. Relational tone will moderate the indirect effect of AI authorship on trust through negative emotions. Specifically, warm-toned apologies will reduce the negative impact of AI authorship by lowering negative emotions, which will, in turn, increase trust and ultimately enhance forgiveness intentions.

H6. Relational tone will moderate the indirect effect of authorship attribution on trust through perceived sincerity and trust. Specifically, warm-toned apologies will increase the perceived sincerity more when the apology is attributed to a human than to AI, which will, in turn, increase trust and ultimately enhance forgiveness intentions.

3. Methods

After receiving IRB approval, we conducted an online experiment via Qualtrics, in which participants were presented with a simulated apology statement from a fictitious watch company, "Novera," addressing a data breach crisis, presented as a post on X (formerly Twitter). A representative U.S. sample of participants who were at least 18 years old was recruited in February 2025 through Prolific Academic. Participants were randomly assigned to one of six conditions in a 3 (authorship attribution: 100 % AI vs. 100 % human vs. control) $\times$ 2 (relational tone: competence vs. warmth) between-subjects factorial design. The control condition was modeled after the baseline control condition employed in Zhang and Gosline's (2023) research. Samples of the stimuli are available in the data repository (Appendix Data, 2025).

3.1. Sample

An a priori power analysis determined that a sample size of at least 251 participants would be required to achieve 95 % power to detect the two-way interaction between authorship attribution and relational tone ($df = 2$), assuming $\alpha = .05$ and a medium effect size ($f = 0.25$). In total, 547 participants completed our study. We excluded 83 participants who provided duplicate responses and did not finish the questions. Thus, the final sample consisted of 464 participants. Respondents had a mean age of 46.2 ($SD = 16.1$) and were relatively evenly divided in terms of self-reported gender (female = 239; male = 216; non-binary = 8; prefer not to say = 1). In terms of ethnicity, the sample consisted of approximately 63.6 % White, 11.2 % Black or African American, 9.1 % Hispanic or Latino, 6.9 % Asian or Asian American, 5.8 % Mixed or Multiracial, 1.7

% American Indian or Alaska Native, 0.6 % Other, 0.4 % Middle Eastern or North African, 0.4 % Native Hawaiian or Pacific Islander, and 0.2 % who preferred not to disclose. Participants received compensation that met Prolific's ethical rewards standards, which ensures fair pay rates per hour.

3.2. Experimental stimuli and procedures

Participants were randomly assigned to one of six experimental conditions and were presented with a simulated apology statement on X from the fictitious watch brand, Novera, regarding a data breach incident. In the apology statement, the organization acknowledged the issue, expressed regret, and outlined the steps it was taking to address the situation and prevent similar incidents in the future. The apology statement was generated by a generative AI model specifically trained to replicate the core elements of Google's February 2024 apology on X. Google issued the original apology in response to inaccuracies in Google Gemini AI's depiction of people of color (Grant, 2024).

To examine the impact of authorship attribution, we used a simulated detection interface modeled after the output style of third-party AI-authorship identification tools. This interface appeared as a static visual element that included a percentage-based estimate indicating the extent to which the apology statement was generated by AI. Unlike prior studies that directly stated the source through text (cf. Lim et al., 2025), our approach conveyed source identity through a visual format that resembled features found in advanced web environments, such as automated labels used for fact-checking on social media. This method provided a contextually grounded way to present authorship cues that reflect how users often receive such information in digital settings. In the 100 % AI-written condition ($N = 153$), participants viewed an icon labeled "AI" accompanied by the message, "100 % AI—We are highly confident this text was generated by AI." In the 100 % human-written condition ($N = 153$), the message appeared with a "Human" icon and the text, "100 % Human—We are highly confident this text is entirely human." The control group ($N = 158$) saw the same apology without any authorship indicator. The full stimuli used in each condition are included in the supplementary materials (see Appendix Data, 2025). Additionally, to manipulate perceptions of relational tone, the tone of the apology statement presented to participants was varied. While the overall structure of the statement remained consistent across conditions, key phrases were modified to emphasize either a warmth ($N = 233$) or competence ($N = 231$) relational tone. Participants took approximately 10 min to complete the survey, and the questionnaire addressed the following measures.

3.3. Measurements

3.3.1. Negative emotions

Negative emotional responses were assessed using items from Mason's (2019) affect scale, which builds on Dillard and Peck's (2001) affective response scale to measure anger and fear. Participants rated the extent to which the company's apology statement made them feel each of the listed negative emotions on a 7-point Likert scale (1 = Not at all, 7 = Extremely). The four items used in the present study were the following: Angry, Annoyed, Irritated, and Aggravated ($M = 2.78$, $SD = 1.53$, $\alpha = .94$).

3.3.2. Perceived sincerity

We used four items adapted from Hu et al. (2021) and Wenzel et al. (2017). Participants evaluated the statement on whether the company's statement seemed sincere, respectful, heartfelt, and genuine. A 7-point Likert scale was used for evaluation, where 1 = strongly disagree, and 7 = strongly agree ($M = 4.52$, $SD = 1.53$, $\alpha = .94$).

3.3.3. Forgiveness intention

Respondents were asked to rate their responses on a 7-point Likert

scale using five items adapted from a forgiveness scale that assessed both relationship maintenance intention and emotional forgiveness (Babin et al., 2021; Tsarenko & Tojib, 2015). Sample statements include, “I will let go of any resentment towards Novera regarding data breach” and “I will forgive Novera for data breach” ($M = 4.40$, $SD = 1.28$, $\alpha = .93$).

3.3.4. Trust in the organization

Trust was measured using a scale adapted from Lim and Jiang (2021), which reflects relational trust—the degree of confidence one party has in the other within an organization-public relationship. This measure captures key sub-dimensions including integrity, dependability, and competence. Participants rated their agreement with eight statements on a 7-point Likert scale such as, “The company keeps its promises” and “The company can be counted on to do what is right” ($M = 4.06$, $SD = 1.25$, $\alpha = .96$).

3.4. Manipulation check

To assess the effectiveness of the authorship attribution manipulation, participants were asked to identify who they believed had written the statement by selecting one of the following options: “fully written by a human,” “fully written by a machine (AI),” or “not sure.” A Chi-square test indicated a significant association between the experimental conditions and participants’ responses, $\chi^2(4) = 243.02$, $p < .001$. To test the manipulation of perceived competence of the apology statement, we asked participants three questions (Fiske et al., 2002) on a 5-point Likert scale, including items such as “It conveys that the company is skillful in managing the issue” and “It conveys that the company is capable of managing the issue” ($M = 3.44$, $SD = 0.96$, $\alpha = .90$). An independent samples *t*-test indicated a significant difference between the two experimental conditions. Participants in the competence relational tone condition perceived the statement as more competent ($M = 3.60$, $SD = 0.86$) than participants in the warmth condition ($M = 3.28$, $SD = 1.02$), $t(462) = -3.66$, $p < .001$. To assess the manipulation of perceived warmth in the apology statement, participants rated four items adapted from Fiske et al. (2002) on a 5-point Likert scale. Sample items included “It conveys that the company is good-natured” and “It conveys that the company is well-intentioned” ($M = 2.92$, $SD = 1.12$, $\alpha = .95$). An independent samples *t*-test revealed a significant difference between the two experimental conditions. Participants in the warmth relational tone condition rated the statement warmer ($M = 3.31$, $SD = 1.11$) than participants in the competence condition ($M = 2.52$, $SD = 0.98$), $t(462) = 8.18$, $p < .001$. Therefore, all three manipulations were successful.

4. Results

H1 predicted that apologies attributed to a human author would generate significantly (H1a) higher perceived sincerity, (H1b) lower negative emotional responses, (H1c) greater trust in the organization, and (H1d) stronger forgiveness intentions compared to those attributed to an AI author. A series of one-way ANOVAs were conducted to examine the effect of authorship attribution on the four dependent variables. As shown in Table 1, authorship attribution had a significant effect on all four outcomes. Participants in the human-attributed apology condition ($M = 4.90$, $SD = 1.33$) reported significantly higher sincerity than those in the AI-attributed condition ($M = 4.11$, $SD = 1.58$), $F(2, 474) = 11.17$, $p < .001$. Post hoc test using Tukey’s *b* showed that the rating in the control group ($M = 4.54$, $SD = 1.56$) fell between the two and was significantly different from the AI condition, but not from the human condition. When it comes to negative emotions, participants exposed to the human-attributed apology ($M = 2.55$, $SD = 1.50$) exhibited significantly lower negative emotions than those who read the AI-attributed version ($M = 3.08$, $SD = 1.54$), $F(2, 474) = 4.93$, $p = .008$. A post hoc test using Tukey’s *b* showed that the control condition ($M = 2.81$, $SD = 1.54$) was not significantly different from either the human or AI conditions. Perceptions of trust also varied by

Table 1
One-way ANOVA results for outcome variables by authorship attribution conditions.

Condition	NE	PS	Trust	FI
	M (SD)	M (SD)	M (SD)	M (SD)
AI ($N = 160$)	3.08 ^a (1.54)	4.11 ^a (1.58)	3.70 ^a (1.26)	4.10 ^a (1.34)
Control ($N = 158$)	2.81 ^{ab} (1.54)	4.54 ^b (1.56)	4.15 ^{ab} (1.20)	4.48 ^{ab} (1.23)
Human ($N = 159$)	2.55 ^b (1.50)	4.90 ^b (1.33)	4.33 ^b (1.21)	4.60 ^b (1.24)
$F(2, 474)$	4.93 ^{**}	11.17 ^{***}	11.16 ^{***}	6.54 ^{**}

Note. Means sharing the same letter are not significantly different at $p < .05$ (Tukey’s *b* test).
* $p < .05$, ** $p < .01$, *** $p < .001$.
NE= Negative Emotions; PS = Perceived Sincerity; FI = Forgiveness Intention.

attribution condition. Participants expressed significantly higher trust in the company when the apology was attributed to a human ($M = 4.33$, $SD = 1.21$) than when it was attributed to AI ($M = 3.70$, $SD = 1.26$), $F(2, 474) = 11.16$, $p < .001$. The post hoc test using Tukey’s *b* revealed that the control condition ($M = 4.15$, $SD = 1.20$) was not significantly different from either. Finally, for forgiveness intentions, participants in the human condition ($M = 4.60$, $SD = 1.24$) showed significantly greater forgiveness than those in the AI condition ($M = 4.10$, $SD = 1.34$), $F(2, 474) = 6.54$, $p = .002$. As with trust, the post hoc test using Tukey’s *b* revealed that the control condition ($M = 4.48$, $SD = 1.23$) was not significantly different from the other two experimental conditions.

Taken together, these results support all proposed hypotheses of H1a, H1b, H1c, and H1d. The ANOVA findings consistently revealed significant differences between the AI and human attribution conditions across all outcome variables, with the human condition generally eliciting more favorable responses. In contrast, the control group (without authorship cues) did not significantly differ from human groups for all dependent variables. These patterns validate the effectiveness of the AI-human attribution manipulation and affirm the distinctiveness of the AI condition, which is the primary focus of this research. Accordingly, subsequent analyses excluded the control group and focused solely on comparisons between the AI and human conditions.

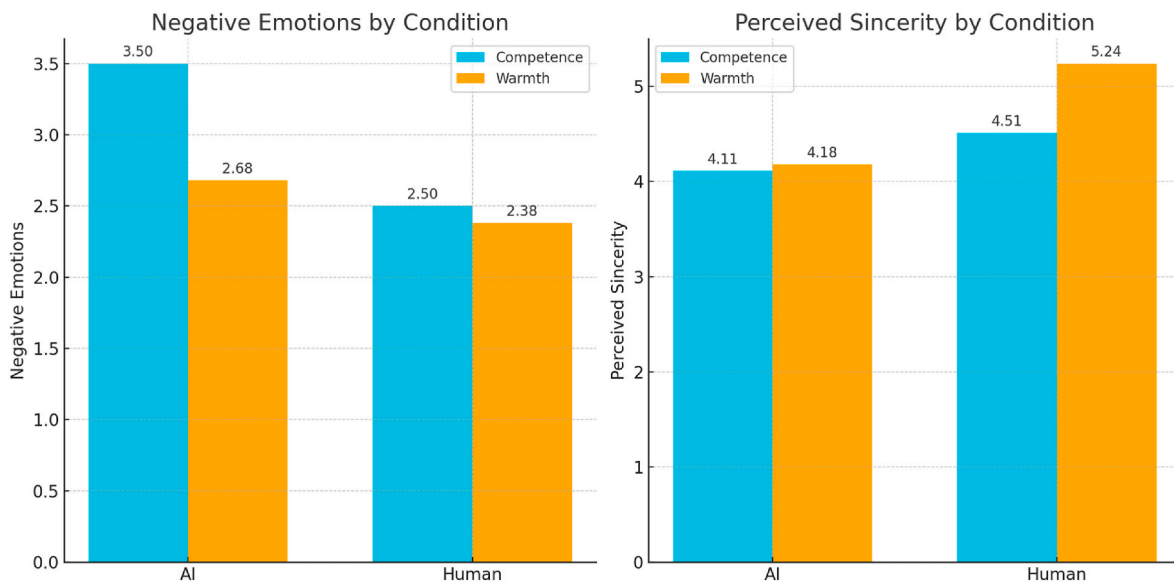
H2 predicted that warm-toned apologies would elicit (H2a) higher perceived sincerity and (H2b) lower negative emotions than competent-toned apologies. To test H2, we conducted two-way ANOVAs using negative emotion and perceived sincerity as the dependent variables, with authorship attribution and relational tone as the two independent variables. Compared to independent samples *t*-tests, this approach allows for the examination of both main and interaction effects of the independent variables. The results in Table 2 revealed a significant main effect of relational tone on negative emotion, $F(1, 302) = 7.81$, $p = .006$, $\eta_p^2 = 0.025$. Participants reported lower negative emotions in the warmth tone condition ($M = 2.53$, $SD = 1.47$) than in the competence tone condition ($M = 3.00$, $SD = 1.55$). Similarly, the ANOVA with perceived sincerity as the dependent variable revealed a significant main effect of relational tone, $F(1, 302) = 5.63$, $p = .018$, $\eta_p^2 = 0.018$. Participants in the warmth tone condition ($M = 4.71$, $SD = 1.61$) rated the apology as more sincere than those in the competence tone condition ($M = 4.31$, $SD = 1.39$). Therefore, both H2a and H2b were supported.

H3 predicted that warm-toned apologies would reduce negative emotions more effectively when the apology was attributed to AI than when attributed to a human. H4 predicted that warm-toned apologies would enhance perceived sincerity more strongly in human-attributed than AI-attributed apologies. Using the same two-way ANOVA models as above, we tested H3 and H4, with negative emotions and perceived sincerity as dependent variables (see Fig. 1). The analysis revealed a significant interaction effect between authorship attribution and relational tone, $F(1, 302) = 4.40$, $p = .037$, $\eta_p^2 = 0.014$. Specifically, participants in the AI-attributed apology condition ($M = 3.09$, $SD = 1.57$)

Table 2

Two-way ANOVA results for outcomes variables as a function of authorship attribution and relational tone.

Dependent variable	Source	SS	df	F	η^2	power
Negative emotions	Authorship attribution	32.21	1	15.02***	0.05	0.97
	Relational tone	16.75	1	7.81**	0.03	0.80
	Authorship attribution x Relational tone	9.43	1	4.40*	0.01	0.55
	Error	647.68	302			
Perceived sincerity	Authorship attribution	40.70	1	19.29***	0.06	0.99
	Relational tone	11.87	1	5.63*	0.02	0.66
	Authorship attribution x Relational tone	8.33	1	3.95*	0.01	0.51
	Error	637.27	302			
Trust in the organization	Authorship attribution	22.64	1	18.30***	0.06	0.99
	Relational tone	3.63	1	2.94	0.01	0.40
	Authorship attribution x Relational tone	0.15	1	0.12	0.00	0.06
	Error	373.62	302			
Forgiveness intention	Authorship attribution	16.26	1	9.82**	0.03	0.88
	Relational tone	5.32	1	3.21	0.01	0.43
	Authorship attribution x Relational tone	0.08	1	0.05	0.00	0.06
	Error	500.28	302			

Note. * $p < .05$, ** $p < .01$, *** $p < .001$.**Fig. 1.** Negative emotions and perceived sincerity by authorship attribution and relational tone.

reported significantly stronger negative emotions compared to those in the human-attributed condition ($M = 2.44$, $SD = 1.41$), $F(1, 302) = 15.02$, $p < .001$, $\eta_p^2 = 0.047$. However, when AI-attributed apologies were presented with a warm tone, warmth significantly reduced negative emotions towards AI-attributed apologies. Specifically, within the AI-attributed condition, a warm tone elicited lower negative emotions ($M = 2.68$, $SD = 1.51$) compared to a competence tone ($M = 3.50$, $SD = 1.53$). This reduction effect was greater in magnitude in the AI-attributed condition ($\Delta M = 0.82$) than in the human-attributed condition, where the difference between warm ($M = 2.38$, $SD = 1.42$) and competent tones ($M = 2.50$, $SD = 1.40$) was substantially smaller ($\Delta M = 0.12$). These results suggest that warmth in tone played a greater role in reducing negative emotional responses when the apology was attributed to an AI source. Therefore, **H3** was supported.

For **H4**, a two-way ANOVA on perceived sincerity revealed a small yet significant interaction effect between attribution and tone, $F(1, 302) = 3.95$, $p = .048$, $\eta_p^2 = 0.013$. Specifically, participants in the human-attributed apology condition ($M = 4.88$, $SD = 1.35$) perceived the statement as more sincere compared to those in the AI-attributed condition ($M = 4.15$, $SD = 1.59$), $F(1, 302) = 19.29$, $p < .001$, $\eta_p^2 = 0.06$. Notably, a warm tone significantly increased perceived sincerity in human-attributed apologies, where perceived sincerity was higher when

message was delivered with a warm tone ($M = 5.24$, $SD = 1.26$) compared to a competent tone ($M = 4.51$, $SD = 1.34$). The human-attributed condition exhibited a larger increase ($\Delta M = 0.73$) compared to the AI-attributed condition, where the difference between warm ($M = 4.18$, $SD = 1.76$) and competent tones ($M = 4.11$, $SD = 1.41$) was relatively small ($\Delta M = 0.07$). These results suggest that warmth in tone played a greater role in enhancing perceived sincerity when the apology was attributed to a human source. Therefore, **H4** was supported.

To test the moderated mediation effects (**H5** and **H6**), we conducted a series of conditional process analyses using SmartPLS. The primary independent variable was authorship attribution, with the human condition serving as the reference category. Relational tone served as a moderator, with competence serving as the reference category. We employed a multiple-mediator model, with negative emotions and perceived sincerity as primary mediators, and trust as an intermediate mediator connecting these variables to the final dependent variable, forgiveness intentions. All indirect effects were estimated using 95 % bias-corrected and accelerated (BCa) bootstrap confidence intervals based on 5000 resamples.

H5 posited that warm-toned apologies would reduce the negative impact of AI authorship by lowering negative emotions, which would, in

turn, increase trust and ultimately enhance forgiveness intentions. The results revealed that relational tone significantly moderated the indirect effect of authorship attribution on forgiveness intention through negative emotion and trust, $\beta = 0.20$, $t = 2.13$, $p < .05$, BCa 95 % CI [0.01, 0.38] (see Table 3).

Results of direct effect show that AI-attributed apologies significantly increased negative emotions ($\beta = 0.66$, $p < .001$), which in turn reduced trust in the organization ($\beta = -0.59$, $p < .001$). Trust, in turn, strongly predicted forgiveness intention ($\beta = 0.72$, $p < .001$). The effect of AI authorship on negative emotions was significantly reduced under the warm tone condition ($\beta = -0.46$, $p = .034$). This suggests that a warm relational tone helped buffer the emotional backlash often triggered by AI-authored messages. Although the conditional indirect effect of AI authorship on forgiveness intention through negative emotions and trust was not statistically significant under the high warmth condition ($\beta = -0.10$, BCa 95 % CI [-0.29, 0.02]), it was significant under the low warmth (competence) condition ($\beta = -0.30$, BCa 95 % CI [-0.51, -0.11]). More importantly, the moderated indirect effect was significant ($\beta = 0.20$, BCa 95 % CI [0.01, 0.38]), which indicates that the strength of the indirect effect differed by relational tone. These results imply that warm tone reduced the negative emotional pathway linking AI authorship to lower forgiveness. Taken together, these results support H5.

H6 posited that warm-toned apologies would increase perceived sincerity more when the apology is attributed to a human than to AI, which would, in turn, increase trust and ultimately enhance forgiveness intentions. According to Table 3, the conditional indirect effect from authorship attribution to forgiveness intention via perceived sincerity

and trust was significant under high warmth ($\beta = -0.39$, $t = 4.06$, $p < .001$, BCa 95 % CI [-0.59, -0.21]). The indirect path from authorship attribution to trust via perceived sincerity was also significant ($\beta = -0.54$, $t = 4.40$, $p < .001$, BCa 95 % CI [-0.78, -0.30]). The moderated indirect effect for the full pathways (Warmth X AI $\rightarrow$ Sinc $\rightarrow$ TRST $\rightarrow$ FI) was statistically significant, $\beta = -0.24$, $t = 1.95$, $p < .05$, BCa 95 % CI = [-0.49, 0.00]. The negative coefficients reflect the dummy-coded comparison where AI was coded as 1 and human as 0. This means that, as attribution shifts from human to AI, the indirect pathway through sincerity and trust decreases. In other words, warm-toned apologies attributed to a human lead to significantly higher perceived sincerity and trust, and ultimately greater forgiveness intentions, than warm-toned apologies attributed to AI. Therefore, H6 is supported.

5. Discussion

This study addressed the central question raised in the introduction: How can AI-generated apologies, despite their perceived deficits in sincerity and empathy, still foster trust and forgiveness? Our findings suggest that under specific conditions, particularly when a warm relational tone is used, AI-attributed crisis messages can achieve important communication goals by reducing emotional resistance. To examine this, we focused on the strategic use of relational tone, comparing warm-toned apologies with competence-toned ones, and explored how these tones interact with authorship attribution (AI vs. human). Below, we summarize our key findings and situate them within the broader body of related research.

First, before examining interaction effects, we tested the main effects

Table 3
Result of process model testing moderated mediation hypotheses.

Direct Effects									
Model 1					Model 2				
Paths	β	SE	t	BCa 95 % CI	Paths	β	SE	t	BCa 95 % CI
AI $\rightarrow$ NEG	0.66	0.15	4.41***	[0.36, 0.94]	AI $\rightarrow$ SINC	-0.26	0.15	1.81	[-0.56, 0.01]
NEG $\rightarrow$ FI	-0.14	0.04	3.12**	[-0.22, -0.05]	SINC $\rightarrow$ FI	0.10	0.06	1.68	[-0.02, 0.23]
NEG $\rightarrow$ TRST	-0.59	0.04	14.47***	[-0.66, -0.50]	SINC $\rightarrow$ TRST	0.77	0.03	30.87***	[0.72, 0.82]
TRST $\rightarrow$ FI	0.72	0.04	19.16***	[0.64, 0.79]	TRST $\rightarrow$ FI	0.72	0.06	12.98***	[0.61, 0.82]
WARMTH $\times$ AI $\rightarrow$ NEG	-0.08	0.15	0.52	[-0.36, 0.22]	WARMTH $\times$ SINC $\rightarrow$ SINC	0.48	0.14	3.54***	[0.21, 0.75]
WARMTH $\times$ AI $\rightarrow$ NEG	-0.46	0.22	2.12*	[-0.89, -0.03]	WARMTH $\times$ AI $\rightarrow$ SINC	-0.44	0.22	1.99*	[-0.85, 0.00]
Conditional Indirect Effects									
Model 1					Model 2				
Paths	β	SE	t	BCa 95 % CI	Paths	β	SE	t	BCa 95 % CI
AI $\rightarrow$ NEG $\rightarrow$ TRST (High WARMTH)	-0.12	0.09	1.24	[-0.31, 0.06]	AI $\rightarrow$ SINC $\rightarrow$ TRST (High WARMTH)	-0.54	0.12	4.40***	[-0.78, -0.30]
AI $\rightarrow$ NEG $\rightarrow$ TRST (Low WARMTH)	-0.39	0.09	4.18***	[-0.57, -0.21]	AI $\rightarrow$ SINC $\rightarrow$ TRST (Low WARMTH)	-0.20	0.11	1.80	[-0.44, 0.01]
AI $\rightarrow$ NEG $\rightarrow$ FI (High WARMTH)	-0.02	0.02	1.20	[-0.08, 0.01]	AI $\rightarrow$ SINC $\rightarrow$ FI (High WARMTH)	-0.07	0.05	1.60	[-0.18, 0.01]
AI $\rightarrow$ NEG $\rightarrow$ FI (Low WARMTH)	-0.09	0.03	2.65**	[-0.17, -0.04]	AI $\rightarrow$ SINC $\rightarrow$ FI (Low WARMTH)	-0.03	0.02	1.15	[-0.10, 0.00]
AI $\rightarrow$ NEG $\rightarrow$ TRST $\rightarrow$ FI (High WARMTH)	-0.08	0.07	1.22	[-0.23, 0.04]	AI $\rightarrow$ SINC $\rightarrow$ TRST $\rightarrow$ FI (High WARMTH)	-0.39	0.10	4.06***	[-0.59, -0.21]
AI $\rightarrow$ NEG $\rightarrow$ TRST $\rightarrow$ FI (Low WARMTH)	-0.28	0.07	4.07***	[-0.42, -0.15]	AI $\rightarrow$ SINC $\rightarrow$ TRST $\rightarrow$ FI (Low WARMTH)	-0.15	0.08	1.79	[-0.31, 0.00]
Moderated Indirect Effect									
Model 1					Model 2				
Paths	β	SE	t	BCa 95 % CI	Paths	β	SE	t	BCa 95 % CI
WARMTH $\times$ AI $\rightarrow$ NEG $\rightarrow$ TRST	0.27	0.13	2.12*	[0.02, 0.53]	WARMTH $\times$ AI $\rightarrow$ SINC $\rightarrow$ TRST	-0.34	0.17	2.00*	[-0.65, 0.00]
WARMTH $\times$ AI $\rightarrow$ NEG $\rightarrow$ FI	0.06	0.04	1.64	[0.01, 0.16]	WARMTH $\times$ AI $\rightarrow$ SINC $\rightarrow$ FI	-0.04	0.04	1.22	[-0.15, 0.00]
WARMTH $\times$ AI $\rightarrow$ NEG $\rightarrow$ TRST $\rightarrow$ FI	0.20	0.09	2.13*	[0.01, 0.38]	WARMTH $\times$ AI $\rightarrow$ SINC $\rightarrow$ TRST $\rightarrow$ FI	-0.24	0.12	1.95*	[-0.49, 0.00]

Note. BCa 95 % CI indicates bias-corrected and accelerated bootstrap (5000 resample) confidence interval. AI: Dummy-coded variable where AI = 1 is AI-attributed source (independent variable). The human condition serves as the reference group (not explicitly coded). WARMTH: Dummy-coded variable where WARMTH = 1 is warmth-focused tone (moderator). The competence-focused tone serves as the reference group. NEG: Perceived negative emotion from the company's statement (mediator). SINC: Perceived sincerity of the company's statement (mediator). TRST: Trust in the company (mediator), FI: Forgiveness intention following the crisis (dependent variable). Low WARM: Indirect effect of AI $\rightarrow$ NEG $\rightarrow$ TRST $\rightarrow$ FI and AI $\rightarrow$ SINC $\rightarrow$ TRST $\rightarrow$ FI when WARMTH = 0 (competence condition). High WARMTH: Indirect effect of AI $\rightarrow$ NEG $\rightarrow$ TRST $\rightarrow$ FI and AI $\rightarrow$ SINC $\rightarrow$ TRST $\rightarrow$ FI when WARMTH = 1 (warmth condition). Moderated Indirect Effect: Difference between conditional indirect effects at low and high levels of WARMTH. * $p < .05$; ** $p < .01$; *** $p < .001$.

of authorship attribution on four core outcomes: perceived sincerity, negative emotions, trust, and forgiveness intentions (H1a–H1d). Across all conditions, human-attributed apologies consistently elicited higher sincerity, trust, and forgiveness, and evoked fewer negative emotions than AI-attributed ones. The Tukey's b post hoc test results (see Table 1) further clarify whether the observed effects were driven by the message content itself or by authorship attribution. Because all participants received the identical apology text, the control group—who received no authorship information—serves as a baseline. The Tukey's b post hoc analyses confirmed statistically significant differences between the AI and human conditions across all key outcomes, with the control group positioned in the middle. These patterns indicate that the main effects resulted primarily from participants' perceptions of who authored the message and the relational tone it conveyed, rather than from the content itself.

In testing H2a and H2b, we also confirmed that warm-toned apologies elicited lower negative emotions and higher perceived sincerity than competence-toned apologies. These findings support prior research that showed the importance of reducing negative emotions in achieving crisis communication outcomes (Barbarossa et al., 2018; Stephens et al., 2019). These results are also aligned with Hu and Pan (2024), who demonstrated that warmth-evoking, informal styles improve forgiveness intentions by enhancing perceptions of warmth. Importantly, warmth appeared to foster more effortful engagement and trust, especially when the crisis demanded a human-centered response (Roccapriore & Pollock, 2023). Our research contributes to the growing literature on human-AI interactions (HAI), particularly regarding emotional and relational outcomes. Prior to the emergence of GenAI, studies reported limitations in AI systems' ability to elicit emotionally resonant responses. For instance, Drouin et al. (2022) found that users saw interactions with AI chatbots, like Replika, as moderately pleasant but rated them lower in responsiveness, liking, and relational warmth than human conversations. However, our findings indicate that recent advances in GenAI may begin to address these limitations. We found that warm-toned AI-authored apologies reduced negative emotions and improved audience evaluations, such that emotionally attuned language helped GenAI messages reduce negative emotions, even in crisis contexts.

After confirming the potential positive effect of warmth in relational tone, we tested whether it help reduce negative emotions for AI-attributed apologies (H3) and further increase perceived sincerity for human-attributed apologies (H4) compared to competence-toned apologies. We found evidence that supports these hypotheses. The potential efficacy of warmth in reducing negative emotions suggests that adopting a warm relational tone can partially compensate for AI's perceived relational deficits. This supports Cuddy et al.'s (2011) insight that organizations perceived as cold, even when they are competent, are more likely to elicit negative emotional responses in high-stakes situations. When warmth is present, it can help reduce public outrage and foster more favorable evaluations, even if the message source does not have fully human characteristics. As previous work has shown, non-humanoid agents are often perceived as less capable of conveying empathy (Choi et al., 2021), which can trigger negative machine heuristics and limit emotional resonance. However, adding warmth, such as through language or tone, may help mitigate these limitations (Cao et al., 2024; Oh & Ki, 2024). Thus, the affective gap between AI and human communicators may be reduced when AI messages present warmth. Additionally, the stronger effect of warmth on sincerity in the human condition aligns with findings by Xiao and Yu (2025), who showed that AI agents often fail to match the sincerity and trustworthiness of human communicators, particularly when empathy is required. Although warmth can simulate human-like qualities (Hu & Pan, 2024), the perceived authenticity of emotional expression remains lower when the communicator is AI (Choi et al., 2021; Kim & Song, 2021). As a result, sincerity is more readily attributed to human apologies, even when both use the same relational tone.

Next, we tested the moderated mediation hypotheses (H5 and H6),

which proposed that warmth moderates the indirect effect of authorship attribution on trust and forgiveness intention by either reducing negative emotions or increasing perceived sincerity. For H5, we predicted that AI-attributed apologies would have a more positive indirect effect on trust and forgiveness when presented in a warm tone rather than a competent tone, due to a greater reduction in negative emotions. Consistent with our prediction, the warm tone moderated the indirect effect of AI-attributed apologies on negative emotions. Specifically, when the apology was delivered in a warm tone, the negative indirect effect of AI authorship on forgiveness intention through heightened negative emotions and reduced trust became weaker. In other words, warmth helped ease the emotional discomfort often associated with AI-generated apologies. This suggests that AI-authored apologies can be perceived more favorably when conveyed in a warm relational tone. This result is consistent with prior studies (Cao et al., 2024; Xiao & Yu, 2025), which found that warmth-inducing cues lowered emotional resistance to AI by enhancing perceptions of authenticity and emotional connection.

In testing H6, we found that when AI-attributed apologies were presented in a warm tone, they still produced a negative indirect effect on trust and forgiveness through reduced perceptions of sincerity. Put differently, as the apology becomes more strongly associated with human authorship, this negative indirect effect is reduced because participants perceive the message as more sincere. The two-way ANOVA results reported in Fig. 1 further support this interpretation by showing higher sincerity ratings for warm-toned apologies when attributed to a human.

Together, these findings offer a response to the broader challenge raised in the introduction: While AI-authored apologies face skepticism due to a lack of perceived sincerity and empathy, our results show that these limitations are not insurmountable. With the strategic use of relational tone, particularly warmth, and clear communication of human oversight, organizations can design AI-assisted apologies that resonate with audiences and promote forgiveness, even in high-stakes crises.

6. Theoretical and practical implications

This study offers several theoretical contributions. First, it provides empirical support that a warm relational tone can reduce the negative emotional responses often associated with AI-attributed apologies in crisis settings. By reaffirming the premise of the Stereotype Content Model (SCM) (Fiske et al., 2007) that relational tone itself can shape an individual's perceptions or judgment, our findings further extend the SCM by addressing a persistent challenge in AI-mediated communication, which concerns the widespread perception that AI lacks empathy and emotional depth. These findings extend the Stereotype Content Model (SCM) (Fiske et al., 2007) by addressing a persistent challenge in AI-mediated communication, which concerns the widespread perception that AI lacks empathy and emotional depth. Due to machine heuristics (Yang & Sundar, 2024), audiences tend to associate AI with competence-based traits such as efficiency and intelligence, but not with warmth-related traits such as sincerity, empathy, or moral intention. This perception limits AI's effectiveness in emotionally sensitive contexts such as public apologies. Our findings suggest that warmth, as defined in the SCM, can function as a compensatory signal that improves the perceived sincerity of AI-generated messages. When AI-authored apologies use a warm tone, they may evoke stronger emotional engagement, which helps reduce relational barriers tied to AI's perceived lack of human emotional capacity.

This study also presents a useful theoretical mechanism for understanding how AI versus human authorship attributions can influence crisis communication outcomes when interacting with relational tone. The findings show that human-authored apologies consistently perform better across all key outcomes, and their advantage is explained by higher levels of perceived sincerity. However, AI-authored apologies can

partly compensate for their limitations by reducing negative emotions. In other words, warm-toned apologies enhance perceived sincerity more effectively when attributed to a human, while for AI-attributed messages, warmth significantly reduces negative emotional responses. These results signify the importance of examining the interaction between relational tone and source attribution when theorizing about AI-mediated communication. This echoes broader conclusions in the literature that tone and authorship interact in complex ways, influencing both emotional and cognitive reactions, and ultimately shaping stakeholder outcomes (e.g., [Xiao & Yu, 2025](#); [Yu & Zhao, 2024](#)).

The findings of this study also offer practical implications for organizations using AI in their public communications. Companies using AI-generated apologies should focus not only on the content but also on the relational tone of the message. Incorporating warmth can help reduce negative emotional responses and increase forgiveness. As AI-generated communication often triggers skepticism or emotional distancing, strategically embedding warmth can serve as a relational buffer. For human spokespersons, expressing warmth and sincerity remains essential to maximizing the effectiveness of an apology. Consistent with prior research ([Hu & Pan, 2024](#); [Xiao & Yu, 2025](#)), we found that human-attributed apologies were perceived as more sincere than those attributed to AI, emphasizing the importance of human authorship in building trust and credibility in high-stakes crises. Therefore, prioritizing human involvement or clearly indicating human oversight in AI-generated messages may help strengthen relational repair. However, our study also highlights that AI apologies can still be effective, especially when they employ a warm tone. While the findings of this study suggest that AI-authored crisis messages can become more effective when emulating warmth and sincerity, this raises important ethical considerations. Specifically, designing AI to replicate human expression, especially in sensitive contexts like apologies, can blur the line between authentic communication and emotional manipulation. For example, [Cao et al. \(2024\)](#) and [Hu and Pan \(2024\)](#) demonstrate that anthropomorphic cues such as warm language and emojis can make AI appear more trustworthy and sincere, but these perceptions are artificially induced. When organizations deploy AI to simulate empathy, they may risk misleading stakeholders into believing that a non-human entity possesses genuine remorse or emotional understanding, even though AI lacks the capacity for authentic emotional experience ([Xiao & Yu, 2025](#)). Practitioners must evaluate ethically problematic practices that may be perceived as performative or deceptive. Transparency about AI involvement may help mitigate concerns while still benefiting from tone-based strategies.

A further managerial implication for using GenAI in crisis communication concerns the importance of prompt engineering in producing effective AI-generated apologies. [Park and Kim \(2023\)](#) demonstrated how GPT-4 could be used to emulate CEO apology statements in the aftermath of corporate crises. They emphasized that the quality and perceived authenticity of these AI-generated messages depend largely on how prompts are constructed. In line with the effectiveness of warm-toned messaging, [Park and Kim \(2023\)](#) also recommended incorporating rhetorical elements such as ethos (credibility) and pathos (empathy) to help AI-crafted apologies resonate more authentically with stakeholders.

Building on this insight, the potential of human-AI collaboration in crafting crisis responses emerges as a meaningful consideration. Rather than viewing AI and human communication as entirely separate, recent studies emphasize the benefits of enhancing professional expertise with GenAI tools. [Park and Kim \(2023\)](#) showed that GPT-4-generated CEO apologies become more effective when communication professionals carefully construct prompts that incorporate rhetorical elements such as credibility and empathy. [Xiao and Yu \(2025\)](#) observed that while AI often lacks relational authenticity, human guidance during message development can reduce this gap. These insights indicate that communication professionals who refine prompts and oversee content generation can help ensure AI-generated messages convey the perception of

sincerity and meet audience expectations. This collaborative model presents a more practical and ethically sensitive way to use GenAI in crisis communication.

7. Limitations and future research

The results of this study point to several limitations commonly associated with experimental research, which should be addressed in future research. First, our experiment focused on a single corporate ability crisis involving a fictional company's data breach. However, crises differ substantially in perceived severity, attributed responsibility, and emotional intensity. For instance, incidents involving product inconvenience (e.g., service delays or minor defects) may be more amenable to AI-mediated apologies, especially when these are carefully constructed using rhetorical prompts that convey warmth and credibility ([Park & Kim, 2023](#)). In contrast, crises involving ethical misconduct, personal harm, or loss of life demand heightened emotional engagement and moral accountability. In such high-emotion contexts, AI-authored messages, even those optimized for warmth, may be seen as performative or lacking genuine remorse ([Piller, 2025](#); [Xiao & Yu, 2025](#)). These conditions may limit the effectiveness of AI-generated apologies and reinforce the need for human presence. To improve generalizability, future research should examine a wider range of crisis types and explore how emotional intensity and contextual framing moderate the reception and effectiveness of AI-generated crisis communication.

Second, while the study successfully manipulated message tone to reflect either relational warmth or competence, real-world apologies addressing data breach scandals are not always written to convey just one of these tones. Instead, such messages often aim to balance or integrate both warmth and competence to enhance effectiveness. To increase external validity, future research should examine how mixed or blended relational tones influence public perceptions and outcomes in more realistic crisis communication contexts.

Third, while this study tested apologies attributed to different agents, an important but underexplored area of research involves moving beyond AI disclosure or detection. Future studies should examine scenarios where actual AI agents function as crisis communicators in real time. This shift would reflect more practical and applied uses of AI in organizational communication.

8. Conclusion

This study examined how AI-authored apologies influence public responses during a corporate ability crisis. While human-attributed messages were generally perceived as more sincere and effective, our findings indicate that AI-generated messages also fostered trust and forgiveness when accompanied by a warm relational tone. Warmth reduced negative emotions, which then increased trust and forgiveness intentions. These results underscore the importance of incorporating a human-like relational tone, such as warmth, to make AI-assisted apologies more effective. With prompt engineering and access to tailored training data, GenAI can be used to produce messages that align with an organization's communicative goals, tone, and brand voice ([Park & Kim, 2023](#)). This makes it a promising tool for generating timely and context-appropriate crisis responses. However, the effectiveness of AI-generated messages likely varies by crisis type and emotional intensity. While our study focused on a data breach scenario, future research should examine more emotionally charged crises where AI may be perceived as less appropriate. Ultimately, our findings underscore that GenAI can support effective crisis communication when paired with human oversight and strategic messaging. Crafting AI responses that incorporate warmth and empathy offers one path toward closing the sincerity gap and improving stakeholder reception in sensitive situations.

CRediT authorship contribution statement

Joon Soo Lim: Writing – original draft, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Nalae Hong:** Writing – original draft, Visualization, Methodology, Formal analysis, Data curation. **Erika Schneider:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.chb.2025.108761>.

Data availability

I have shared the link to my data.

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